TEST REPORT

Summary and closure of the testing: web shop on Kubernetes

Project / system	Web shop on Kubernetes: four Python/Flask microservices with SQL Server
Release / version	1.2.0 (build 1.2.0-fcde5e3ea47f); previous release 1.1.0
Test period	2026-11-07 to 2026-11-14
Document owner	Jakub Samborski
Date	2026-11-14
Status	Final

Test cases, test results and traceability to the requirements are in docs/v1.2/test-management.xlsx. The plan is TP-WEBSHOP-1.2 (docs/v1.2/test-plan.pdf). The automated tests are in tests/ and are run with python run.py.

1. Summary

This report summarizes the testing of release 1.2 of the web shop. Release 1.2 adds free delivery to the door from $100.00, product cards with images, a product page with a description, a placeholder for missing images, and an assortment of 20 products. Testing covered the 6 new requirements and a full regression of the 41 requirements from 1.1: 118 test cases at four levels (unit, integration, end-to-end and static).

Result: All 118 test cases passed: 116 automated and 2 manual reviews. The automated suite of 231 tests passed in a full run from an empty environment, and all 47 requirements are covered and passed. Deploying 1.2 over a 1.1 database keeps all data (IT-044).

Defects: No product defects were found. One Low defect in the test environment is open (#9: a request through the port-forward is sometimes reset); it passed in the release run. One minor finding in the image credits was fixed.

Remaining risk: As in 1.1 (no load testing, one environment, not hardened for production), plus the delivery fee that is shown on the receipt but not stored with the order.

Recommendation: GO for release 1.2.0 as a local demonstration system. Not for production use without the actions in section 10.

2. Scope

Testing covered the whole system as deployed by deploy.py: the new requirements REQ-42 to REQ-47 and all requirements from 1.1 as regression. The priority of an area is the highest priority among its requirements.

Area / flow	Test level / test type	Priority	Comment
Delivery fee (REQ-42)	Unit, E2E / functional	High	Boundary values $99.99, $100.00 and $100.01 with door delivery and pickup; the fee shown in the cart and on the receipt; exact decimal amounts.
Product presentation (REQ-43 to REQ-45)	Unit, E2E / functional	High	Cards with image, name, price and stock; product page with a larger image and a description; placeholder for a missing, empty and broken image.
Product data and assortment (REQ-46)	Integration, E2E / functional, data migration	Medium	20 products with description and image; every image can be loaded; upgrade from a 1.1 database keeps users, orders and stock and renames the products.
Image licences (REQ-47)	Static / review	Low	Every image listed in CREDITS.md with source, author and licence; licence allowed; size at most about 200 KB.
Regression of release 1.1 (REQ-01 to REQ-41)	Unit, integration, E2E, static	Critical	All 103 test cases from 1.1: checkout and orders, accounts, catalogue and cart, security, operations and resilience.

Out of scope: Load and performance testing, browsers other than Chromium, visual design and image quality, filters, search and sorting, account lockout, real payment and e-mail providers, and testing in a production environment.

3. Tests executed

Every planned test case was executed. The automated tests run in one pytest run; the two static reviews were done by hand.

Test area	Planned	Executed	Status / comment
Unit tests (UT)	46	46	125 automated tests, all passed. 6 new test cases (UT-041 to UT-046).
Integration tests (IT)	44	44	80 automated tests against the deployed system, all passed, including the upgrade from a 1.1 database. 3 new (IT-042 to IT-044).
E2E tests (E2E)	26	26	26 Playwright tests in Chromium, all passed. 5 new (E2E-022 to E2E-026).
Static review (ST)	2	2	Credentials in the repository (ST-001) and image sources and licences (ST-002). Both passed.
Total	118	118	100 % executed: 231 automated tests and 2 manual reviews.

Requirement coverage: All 47 requirements have at least one test case. The delivery fee, which concerns money, is tested at unit level (boundary values) and end to end (cart and receipt).

Test design techniques: boundary value analysis ($100 limit), equivalence partitioning (image present, missing, empty, broken), data migration test (1.1 database), negative tests and regression.

4. Results

Measure	Result	Comment
Test cases executed	100 % (118 of 118)	116 automated, 2 manual reviews.
Pass rate	100 % (118 of 118)	No failed or blocked test cases in the release run.
Automated tests	231 of 231 passed	Unit 125, integration 80, E2E 26. Run time about 6 minutes.
Requirements	47 of 47 passed	41 from 1.1 (regression) and 6 new.
New test cases	15 of 15 passed	Each was written before the code and seen to fail first.

Measure	Result	Comment
Regression (test cases from 1.1)	103 of 103 passed	7 updated: E2E-008 for 20 products, 6 for the new product names.
Run in a clean environment	Passed	From an empty cluster: new secrets, new database, deploy, all tests, cleanup (2026-11-14).
Open Critical / High defects	0	One Low defect open (#9, test environment).

Analysis: A pass rate of 100 % means that version 1.2.0 behaves as the 118 test cases specify. It does not mean the system is free of defects. Because every new test case was written first and failed before the change was made, the new tests are known to detect the missing behaviour. The upgrade test (IT-044) puts the database back in its 1.1 state before deploying, so the release is also tested against existing data. The results are limited to one environment and say nothing about behaviour under load (section 8).

5. Defects

No product defects were found while testing release 1.2. Defects and findings recorded during the testing:

Severity	Total	Open	Comment / impact
Critical	0	0	-
High	0	0	-
Medium	0	0	-
Low	2	1	Open: #9, IT-025 fails now and then because the port-forward resets the connection (Windows error 10053/10054). 120 of 120 requests sent straight to the container were answered correctly, so the shop is not at fault. It failed in 2 of 3 runs on 2026-11-09 and passed in every run since, including the release run. Fixed: ST-002 found that one image author asks for a specific credit text; CREDITS.md was corrected.

Test code: One defect in the test code was found and fixed during the testing: a new test helper was not exported, so E2E-025 could not start in one run. Known limitation (not a defect): the delivery fee is shown in the cart and on the receipt, but it is not stored with the order.

6. Deviations from plan

The deviations below changed how the testing was done; none of them reduced the scope.

Deviation	Impact on testing	Action / handling
REQ-47 changed: free CC0	The plan's risk action 'only CC0 or	REQ-47 now allows CC0, CC BY, CC BY-

Deviation	Impact on testing	Action / handling
images could not be found for most products	own images' and the expected result of ST-002 changed.	SA, Free Art License, Pexels and Unsplash licences. Every image is credited in CREDITS.md with author, licence and the changes made. ST-002 updated and passed.
Brand names replaced with generic product names; Cat Ears headband became Bunny Ears Headband	7 test cases and their tests changed (UT-014, E2E-009, E2E-012, E2E-013, E2E-017, E2E-018, E2E-024).	Updated in the same pull request as the data. Only products that still have the old name are renamed; old orders keep the name they were bought with (REQ-26). IT-044 checks the renames.
E2E-024 moved from step 4 (product cards) to step 5 (images and assortment)	It needs a product with a real image and description, which only exist after step 5.	Executed in step 5. Only the product name in the test case changed.
IT-025 fails now and then (#9)	A run could fail although the shop works.	Decision before the release run: record it as Blocked if it fails and report it before the go/no-go decision. It passed in the release run.

7. Risks

Risks from the test plan and from the testing, and how they were handled.

Risk	Consequence	Action	Status
Rounding error at the $100 limit	Wrong delivery fee charged.	Boundary value tests with exact decimal amounts at unit and E2E level.	Handled
Changed product data breaks existing flows	Checkout or orders fail.	New fields are optional; full regression on every pull request; upgrade test from a 1.1 database (IT-044).	Handled
Images that are not free to use	Legal problem when published.	Sources and licences in CREDITS.md, reviewed in ST-002 (REQ-47 changed, section 6).	Handled
Tests that depend on the number of products	False failures.	E2E-008 updated in the same pull request as the data change.	Handled
Two branches change the workbook at once	Lost changes (binary file).	One branch at a time.	Handled
Large images	Slow pages and slower tests.	Size limit checked: the largest image is 197 KB.	Handled
Intermittent failure in the	A real failure could be	Investigated (requests straight to the	Open,

Risk	Consequence	Action	Status
test environment (#9)	dismissed as noise.	container) and recorded; every failure is analysed before a re-run.	monitored

8. Remaining risk

Risks and uncertainties that remain after the testing.

Remaining risk	Consequence	Risk level	Risk owner / accepted by
No load or performance testing	Response times and stability under real load are unknown.	Medium	Jakub (project owner)
Not hardened for production: no HTTPS, self-signed database certificate, one shared token between the services	Acceptable on a local machine, not on a public server.	Medium (High if deployed publicly)	Jakub (project owner)
Delivery fee not stored with the order	Order history and order totals in order-service do not include the fee; only the receipt shows it.	Low (Medium if orders are used for accounting)	Jakub (project owner)
IT-025 intermittent in the test environment (#9)	A real failure in that area could be missed.	Low	Jakub (project owner)
Images from third parties	Licence terms or credits could be challenged; some photos show brand logos.	Low	Jakub (project owner)
Tested in one environment: Docker Desktop (kind) and Chromium	Other clusters or browsers may behave differently.	Low	Jakub (project owner)
A product deleted by hand comes back on the next deploy	The database script adds missing products by id.	Low	Jakub (project owner)

9. Conclusion

Testing shows that version 1.2.0 meets all 47 requirements in the tested environment. Door delivery is free from exactly $100.00, products are shown as cards with their images, a missing or broken image shows a placeholder, and all 20 products have a description and an image with a documented source. Everything that worked in 1.1 still works, and an existing 1.1 database is upgraded without losing data. All exit criteria of TP-WEBSHOP-1.2 are met. The main limitations are the missing load test and the delivery fee that is not stored with the order.

Exit criterion	Status	Comment
All new and all regression test cases executed	Met	118 of 118 (15 new, 103 regression).
No open Critical or High defects	Met	0 open. One Low defect open (#9).
Every requirement covered by a passed test case	Met	47 of 47.
The full automated suite passes in a clean environment	Met	231 of 231, python run.py from an empty cluster.
Test report for 1.2 written	Met	This report.

10. Recommendation

Recommendation	GO for release 1.2.0 as a local demonstration system
Motivation	All exit criteria are met, no Critical, High or Medium defects are open, and the release run from an empty environment passed. The open defect (#9) is in the test environment, not in the shop. The remaining risks are documented and are Low or Medium for the intended use.
Conditions / actions	Fix #9 in the test tooling in the next release. Store the delivery fee with the order before order data is used for accounting. Before any production use: the same actions as in the 1.1 report (load testing, HTTPS, a separate identity per service, real payment and e-mail, a shared store for the rate limit).
Responsibility / follow-up	Jakub (project owner), 2026-11-14

11. Approval / recipients

Roles that have read or approved this report.

Role	Name	Date	Comment
Tester / report author	Jakub	2026-11-14	Report owner